BEV HMI Board — STM32H753ZI Resources and Architecture

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Pitch

This board is the HMI and cluster controller for a BEV truck: one STM32H753ZI running Apache NuttX that sits on the vehicle network, drives the displays and keeps its own storage, positioning and cellular link.

It grows out of a validated Nucleo-H753ZI configuration (Ethernet, ST7796 display with touch, OLED, RFID, FDCAN, 11 buttons) and moves it to a custom 4-layer PCB. The office Ethernet PHY of the Nucleo is replaced by automotive 100BASE-T1, and the pin map was re-audited so every peripheral, used or planned, has its own pin with no conflict.

Large screens are delegated: a 10-inch ESP32-S3 display module renders its own UI and receives data over a dedicated SPI link, so the H7 does not need external SDRAM or LTDC.

Scope

Rev A carries every resource below on one LQFP144 MCU, with no pin shared between two functions.

Goals:

- Automotive Ethernet 100BASE-T1 (TI DP83TC812) with TC10 sleep and wake.
- Two CAN FD buses (FDCAN1, FDCAN2) with 24 V fault-protected transceivers.
- 24 V vehicle supply protected to 60 V (ISO 16750-2 suppressed load dump), reverse battery protection, and operation through engine cranking.
- 3.5-inch ST7796 touch display, SSD1306 OLED and MFRC522 RFID, as on the Nucleo build.
- SPI4 link to a 10-inch ESP32-S3 display module, running vrm_net.
- 16 MB QSPI NOR flash for OTA image buffer, logs, configuration and certificates.
- LTE Cat 4 modem (Quectel EC200A-EU) and a separate GNSS receiver.
- IMU (LSM6DSOX, as in MAIA), I2S audio, 1-Wire, an extra I2C bus, 3 ADC inputs and one ADC/DAC pin.
- 11 buttons, 3 LEDs, and one STDC14 header for debug, flash and console.

Non-goals for rev A:

- Gigabit Ethernet: the H753 MAC is 10/100 only (MII/RMII). The PHY footprint is pin-compatible with the 1000BASE-T1 DP83TG720 for a future MCU.
- SD card and a rotary encoder: no conflict-free pins remain in LQFP144 (see Backlog). USB is left out by choice; PA9 to PA12 stay free on the custom PCB.
- Native RGB display with SDRAM: needs LQFP176/208.
- ISO 26262 safety case: the design is a prototype, not a homologated ECU.

Feature list

All 40 features planned for rev A, with their pins and how settled each one is; the last group (B1 to B5) lists what stays out of rev A and why.

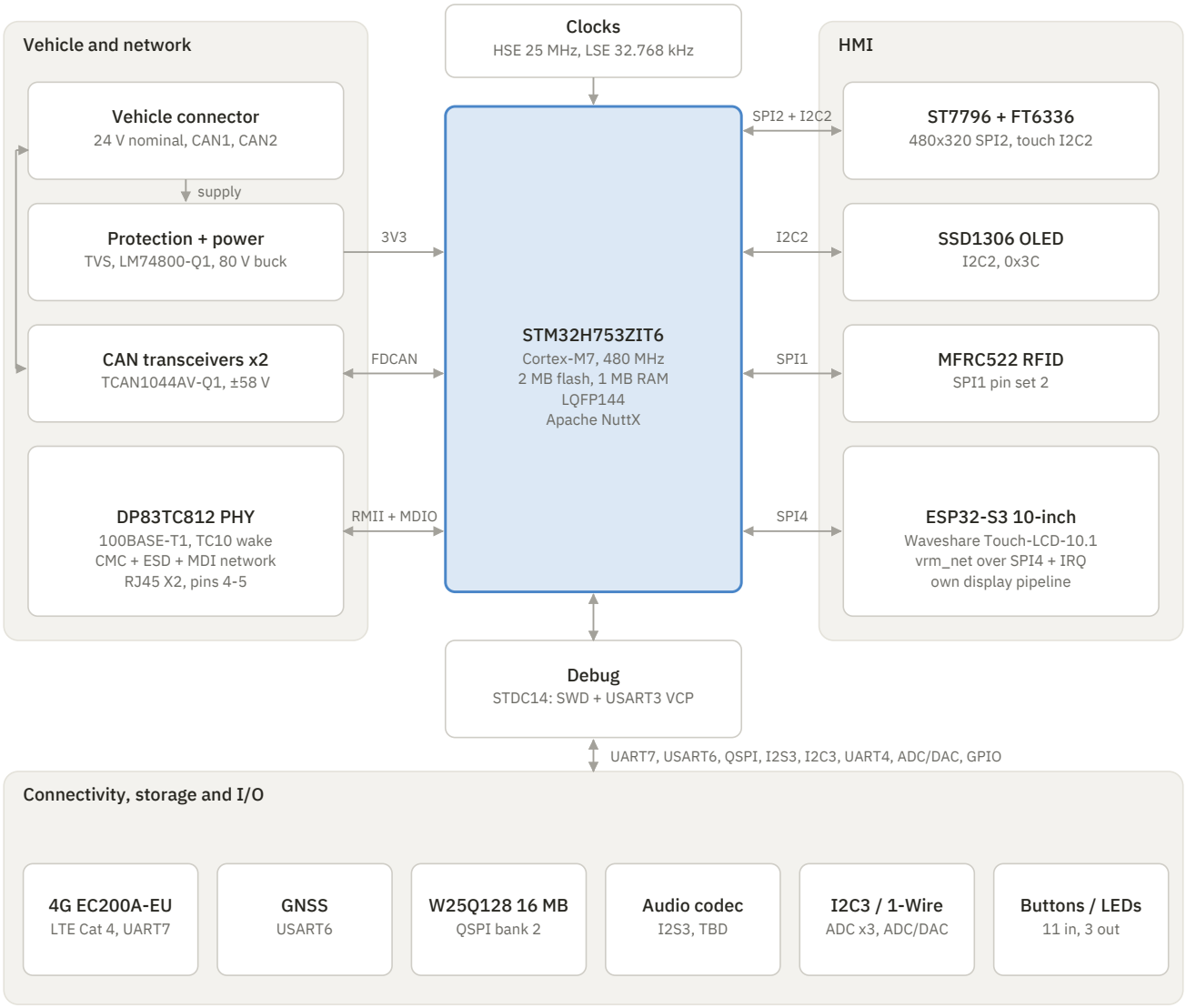
#	Feature	Implementation	Pins	Status
Core				
1	MCU	STM32H753ZIT6, Cortex-M7 480 MHz, 2 MB flash, 1 MB RAM, Apache NuttX	LQFP144	Defined
2	Main clock	HSE 25 MHz crystal	PH0, PH1	Defined
3	RTC clock	LSE 32.768 kHz crystal	PC14, PC15	Defined
4	Power input	24 V nominal, protected to 60 V: fuse, TVS, LM74800-Q1 (reverse battery, overvoltage cut-off, inrush), 80 V buck to 5 V, then 3.3 V; core from internal LDO	VDD, VCAP 71/106	Defined, parts to finalize
5	Cranking	Keeps running during engine start: 80 V buck regulates down to 4.2 V input	-	Defined
6	Ignition input	KL15 through divider, RC and TVS into an ADC pin; polled	PF13 (ADC2)	Defined
Vehicle network				
7	Automotive Ethernet	TI DP83TC812RRHARQ1, 100BASE-T1, RMII master (50 MHz REF_CLK from PHY)	PA1, PA2, PA7, PB13, PC1, PC4, PC5, PG11, PG13	Defined
8	PHY control	Hardware reset and interrupt (EXTI2), TC10 sleep and wake	PG2, PF2	Defined
9	T1 media network	CMC, DC block rated for short to battery, termination, ESD (TI Table 8-1)	-	Defined
10	T1 connector	Inside X2: RJ45 pins 4-5 (T568B pair 1) on a ready-made Cat5e patch cord. Production: MATEnet or H-MTD	X2	Defined
11	CAN FD 1	FDCAN1 + TCAN1044AV-Q1 (±58 V bus fault, VIO 3.3 V) + ESD2CAN24-Q1	PB8, PB9; PD4 STB	Parts proposed
12	CAN FD 2	FDCAN2 + TCAN1044AV-Q1 + ESD2CAN24-Q1	PB6, PB12; PD6 STB	Parts proposed
13	Network protocol	vrm_net over UDP (Ethernet) and over SPI (ESP32-S3 link)	-	Software
HMI				
14	Main display	ST7796 480x320 on SPI2 (pin set 3)	PD3, PC2, PC3; PA3 DC, PA4 CS, PD7 RESET	Defined
15	Backlight	On/off GPIO, no PWM on this pin	PE15	Defined
16	Touch	FT6336 on I2C2 (0x38)	PF0, PF1; PG5 RST, PG8 INT	Defined
17	Secondary display	SSD1306 OLED on I2C2 (0x3C)	PF0, PF1	Defined
18	RFID	MFRC522 on SPI1 pin set 2, up to 10 Mbit/s, on X4	PB3, PB4, PB5; PA0 CS; PG0 RST; PG1 IRQ	Defined
19	10-inch display	Waveshare ESP32-S3-Touch-LCD-10.1; vrm_net over SPI4, STM32 master, 30-40 cm cable	PE12, PE13, PE14; PE0 CS; PD14 IRQ; PD13 EN	SPI defined, framing open

#	Feature	Implementation	Pins	Status
20	Buttons	11 inputs: USER on the board + 10 on X4, closing to 3.3 V	PC13, PB1, PD0, PG4, PD5, PE6, PE3, PD12, PF9, PD15, PE11	Defined
21	LEDs	3 outputs	PB0, PE1, PB14	Defined
Connectivity				
22	Cellular	Quectel EC200A-EU, LTE Cat 4, on UART7; own buck of at least 3 A	PF7, PF6; PF10 PWRKEY, PF11 RESET, PF12 STATUS, PG10 RI	Defined
23	SIM	Nano-SIM holder on the modem	-	Defined
24	LTE antenna	Multiband LTE antenna on u.FL	-	Part open
25	GNSS	GNSS receiver on USART6	PG14, PG9; PG15 RESET, PB7 PPS	Module open
26	GNSS antenna	Active patch antenna on its own u.FL	-	Part open
Storage and update				
27	External flash	Winbond W25Q128JVSIQ, 16 MB, QSPI bank 2	PB2, PC11, PE7, PE8, PE9, PE10	Defined
28	OTA update	New image downloaded over 4G or Ethernet, staged in QSPI flash	-	Bootloader open
29	Logs, config, certificates	LittleFS on the QSPI flash	-	Software
Audio, sensors and I/O				
30	Audio	I2S3 (SPI3 block) + codec + 3.5 mm jack	PC10, PA15, PC12, PC7	Codec open
31	1-Wire	UART4 TX in half-duplex, DS18B20 class sensors	PD1	Driver port
32	Expansion I2C	I2C3, shared with the on-board IMU; optional VL53L5CX ToF for cabin use	PA8, PC9	Defined
33	IMU	ST LSM6DSOXTR (LCSC C481766), 3-axis accelerometer + gyroscope, the same part as MAIA; I2C3 at 0x6A	PA8, PC9; PG3 INT1 (polled)	Defined
34	Analog inputs	3 ADC channels	PC0, PF3, PF4	Defined
35	Analog in or out	ADC12_INP19 or DAC1_OUT2	PA5	Defined
Debug				
36	Debug and flash	STDC14 header (2x7, 1.27 mm) for an external ST-LINK-V3: SWD debug and flash; SWO and JTAG not wired	PA13, PA14, NRST	Defined
37	Serial console	USART3 (NSH) on the same STDC14; the ST-LINK-V3 shows it on the PC as a USB serial port (VCP)	PD8, PD9	Defined
38	USB service port	None in rev A. Possible later: native USB-C on OTG_FS for DFU flashing (ROM bootloader, BOOT0) and a CDC-ACM console; debug still needs SWD	PA11, PA12 (free)	Open decision
Board				

#	Feature	Implementation	Pins	Status
39	PCB	110 x 130 mm, 4-layer, slides into an extruded aluminium enclosure; vehicle connectors on one end plate, panel connectors on the other	-	Concept
40	Connectors	Prototype families, no crimping: X1 DB15 (solder), X2 RJ45 (ready-made cable), X3 and X4 IDC box headers (ribbon pressed in a vise), X5 box header, STDC14, RF via u.FL pigtails to SMA bulkheads	-	Defined, codes to confirm
Not in rev A				
B1	microSD	SDMMC1	PC9-PC12 taken	Backlog
B2	Rotary encoder	Timer encoder mode	No free CH1+CH2 pair	Backlog
B3	USB OTG	Device or host	PA9-PA12 free on the custom PCB	Not planned
B4	7-inch RGB display	LTDC + 16-bit SDRAM on FMC	Needs LQFP176/208	Backlog
B5	Gigabit Ethernet	1000BASE-T1, DP83TG720 pin-compatible	Needs an RGMII MCU	Backlog

System overview

One MCU, three sides: vehicle network on the left, HMI on the right, services below



system overview · buses, power and partitioning

The MCU sits between the vehicle side and the HMI side. Ethernet reaches it as RMII from the T1 PHY, which also generates the 50 MHz reference clock. Both displays, the RFID reader and the ESP32-S3 module hang on separate SPI buses, so a slow device never blocks the main screen. Everything else, from the modem to the buttons, uses its own UART, QSPI, I2S, I2C, ADC or GPIO pin.

Board outline and enclosure

A rectangle of 110 x 130 mm fits the parts best: the board slides into a cheap extruded aluminium box, and each end plate carries one group of connectors.

- Why a rectangle: modules, connectors and the LQFP144 are rectangular, so a rectangle packs them without dead corners and panelizes well at JLCPCB. A circle wastes the corners and needs a custom housing; a square would push connectors onto three or four sides.
- Why connectors on the two short ends: an extruded profile is open at both ends, so all cable openings sit on two flat end plates. The board slides into the rails of the profile and needs no mounting holes.
- Zones along the length: vehicle end (X1, X2, protection, bucks, CAN, T1, modem, RF bulkheads), MCU core in the middle, panel end (X4, X3, debug, GNSS, LEDs). Supply current and switching noise stay at the vehicle end.
- The aluminium profile is the chassis ground and spreads the heat of the MCU, the modem and the bucks; the end plates are bonded to it.
- The display case holds the ST7796 3.5-inch panel and the SSD1306 0.96-inch OLED and is fed by the X4 ribbon. In the render the ribbon is drawn at about 36 cm, the length planned for the panel cable.



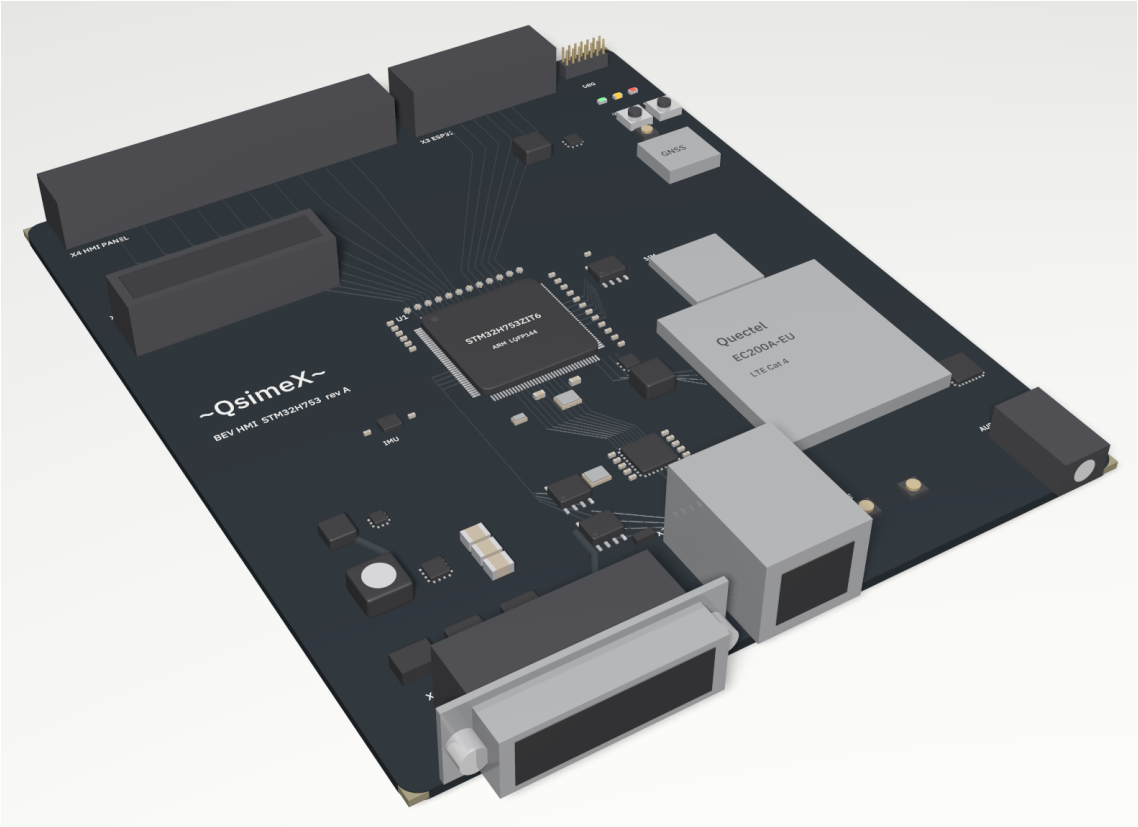
system render \u00b7 ECU enclosure, display case and X4 harness at the same scale \u00b7 POV-Ray

Item	Size (approximate)	Notes
PCB	110 x 130 x 1.6 mm, 4-layer	Rails on both long edges
ECU enclosure	About 120 x 40 x 140 mm with end plates	Extruded aluminium, black anodized end plates
Vehicle end plate	X1 DB15, X2 RJ45, 3 SMA, audio jack	Faces the vehicle harness
Panel end plate	X4 IDC 2x20, X3 IDC 2x8, STDC14, 3 light pipes	Faces the cab panel

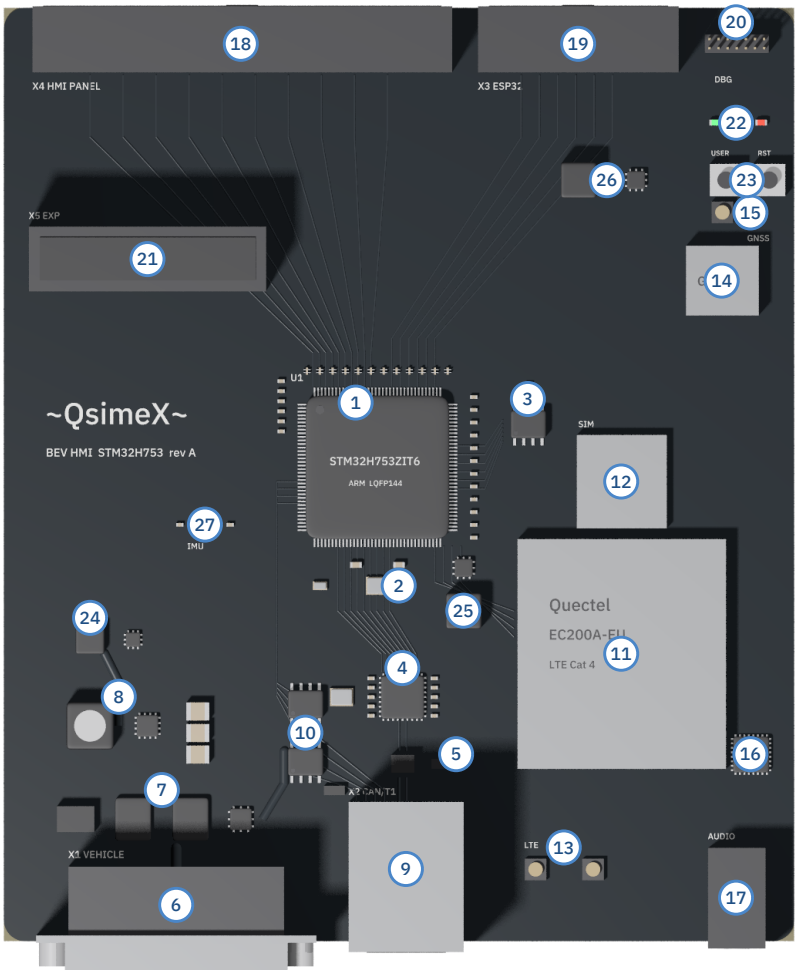
Item	Size (approximate)	Notes
Display case	About 150 x 80 x 24 mm	ST7796 active area about 74 x 49 mm, SSD1306 about 22 x 11 mm; confirm with the chosen modules
X4 ribbon	40-way, about 36 cm	Grey ribbon, red edge on pin 1

Concept render

The board itself, 110 x 130 mm: vehicle end at the front of the render (bottom of the top view), panel end at the back.



concept render · POV-Ray · 110 x 130 mm, 4-layer, black solder mask, ENIG. Not a placement study



top view · numbers refer to the table on the next page

Placement

The placement follows three rules: noise sources away from sensitive inputs, every cable at an edge, and the shortest possible high-speed paths.

#	Block	Position and reason
1	STM32H753ZIT6	Centre; decoupling on all four sides
2	HSE 25 MHz crystal	Next to PH0/PH1 with a ground guard
3	W25Q128 QSPI flash	Right of the MCU, short bank-2 traces
4	DP83TC812 PHY	Between the MCU and X2; RMII kept short
5	T1 front end	CMC, ESD and DC block right behind the RJ45
6	X1 DB15	Vehicle end plate
7	Protection	TVS, two MOSFETs and LM74800-Q1 right behind X1
8	5 V buck (80 V)	Vehicle end, small switching loop
9	X2 RJ45	Vehicle end plate, CAN x2 + T1
10	CAN transceivers	Between X1 and X2, ESD at the jack
11	4G module EC200A-EU	Vehicle end, short pigtails to the SMA jacks
12	Nano-SIM holder	Next to the modem
13	LTE u.FL (main, diversity)	Pigtails to SMA on the vehicle plate
14	GNSS module	Panel end, far from the LTE module
15	GNSS u.FL	Pigtail to SMA on the vehicle plate
16	Audio codec	Next to the jack, away from the buck
17	Audio jack	Vehicle end plate
18	X4 IDC 2x20	Panel end plate, ribbon to the display case
19	X3 IDC 2x8	Panel end plate, ribbon to the ESP32-S3
20	STDC14 debug	Panel end plate, reachable without opening
21	X5 expansion	Inside, 2x10 box header
22	LEDs	Panel end, light pipes in the plate
23	USER and RESET	Panel end, inside
24	3.3 V buck	Next to the 5 V buck
25	Modem buck (3 A)	Next to the modem supply pins
26	ESP32 5 V buck	Next to X3
27	IMU LSM6DSOX	Near the board centre, rigid, away from the bucks

Layout rules:

- Stack-up: signal / ground / power / signal, with one continuous ground plane and no splits.
- The MDI pair runs as a controlled differential pair from the PHY through the CMC to the connector; values from the DP83TC812 data sheet, layout per TI EMC guidance (SNLA434).

- The 50 MHz REF_CLK and the RMII lines stay away from the RF corner.
- The LTE and GNSS antennas sit on different edges; the modem supply has its own bulk capacitance for transmit bursts.
- Zones along the length: vehicle end (X1, protection, bucks, CAN, T1, modem, RF), MCU core in the middle, panel end (X4, X3, debug, GNSS). Noise and current flow one way and never cross the MCU.

Hardware architecture

Every subsystem below has a pin in the rev27 map; parts marked "not chosen" still need a selection.

Subsystem	Choice	Notes
MCU	STM32H753ZIT6, LQFP144	Cortex-M7 up to 480 MHz, 2 MB flash, 1 MB RAM; 14 GPIOs still free after the rev27 plan (all on X5), plus the 6 Nucleo-only USB pins on the custom PCB; all 16 EXTI lines in use
Clocks	HSE 25 MHz crystal (PH0/PH1); LSE 32.768 kHz (PC14/PC15, optional)	The PHY has its own 25 MHz crystal
Power	24 V nominal, protected to 60 V; LM74800-Q1 front end, LMR38020 (80 V, 2 A) to 5 V, then 3.3 V	Core from the internal LDO (VCAP pins 71, 106); see Power and protection
Automotive Ethernet	TI DP83TC812RRHARQ1 (LCSC C3225810)	100BASE-T1, RMII master: 50 MHz REF_CLK from RX_D3 to PA1; TC10; pin-compatible with DP83TG720
T1 media network	CMC 200 µH, DC block 0.1 µF, CM termination 1 kΩ, MDI coupling 4.7 nF, ESD shunt 100 kΩ	From DP83TC812 data sheet Table 8-1; DC block caps rated well above the battery (100 V class) to survive a short
T1 connector	RJ45 X2, pins 4-5 (T568B pair 1)	Twisted pair all the way to the jack; jack without magnetics; MATEnet or H-MTD for production
PHY supplies	VDDIO, VDDMAC, VDDA, VSLEEP at 3.3 V	VSLEEP from an always-on low-Iq LDO (TPS7B81-Q1 per TI) if TC10 is used
CAN	FDCAN1 (PB8/PB9), FDCAN2 (PB6/PB12)	TCAN1044AV-Q1: ±58 V bus fault, VCC 5 V, VIO 3.3 V, standby on PD4/PD6 (proposed, check LCSC stock)
Main display	ST7796 480x320, SPI2	FT6336 touch on I2C2; backlight PE15 is on/off only
Secondary display	SSD1306 OLED, I2C2	Address 0x3C; touch is 0x38 on the same bus
RFID	MFRC522, SPI1 pin set 2	CS on PA0, RST on PG0, IRQ on PG1 (polled); SPI limited to 10 Mbit/s; PB3 is SWO, so no SWO trace
10-inch display	ESP32-S3-Touch-LCD-10.1, SPI4	STM32 is master, CS on PE0, data-ready IRQ on PD14, enable on PD13; vrm_net framing with CRC
Storage	Winbond W25Q128JVSIQ, 16 MB, QSPI bank 2	OTA buffer, logs, config, certificates
Cellular	Quectel EC200A-EU, LTE Cat 4, UART7	Supply 3.4 to 4.5 V (3.8 V typ.), at least 2.8 A capability; PWRKEY, RESET, STATUS, RI on PF10, PF11, PF12, PG10; nano-SIM; LTE antenna on u.FL
GNSS	Module not chosen, USART6	Reset on PG15, PPS on PB7; active patch antenna on its own u.FL
Audio	I2S3 (CK, WS, SDO, MCK)	Codec not chosen; the SPI3 block is used as I2S3, so SPI3 stays disabled in Kconfig

Subsystem	Choice	Notes
1-Wire	UART4_TX half-duplex (PD1)	DS18B20 class sensors
Expansion I2C	I2C3 (PA8/PC9)	Shared with the IMU; for example a VL53L5CX ToF sensor
IMU	LSM6DSOXTR (LCSC C481766) on I2C3, address 0x6A	INT1 on PG3, polled (EXTI3 is BTN6); mounted rigidly near the board centre, away from the bucks
Analog	ADC on PC0, PF3, PF4; ADC/DAC on PA5	Replaces the rotary encoder, which has no free timer pair
Inputs and outputs	11 buttons, 3 LEDs	USER on PC13 plus 10 external buttons
Debug	STDC14 header + external ST-LINK-V3	One probe gives debug, flash and USB serial (VCP on USART3, PD8/PD9), as on the Nucleo; SWO (PB3) and JTAG (PA15, PB4) stay unwired because those pins are in use

Power and protection

The board runs from the 24 V truck supply, survives the suppressed load dump of a 24 V system (58 V, ISO 16750-2 test B) with a 60 V design target, and keeps working while the engine cranks.

Stage	Part	Purpose
1 Fuse	Slow-blow, sized above the cranking current and the controlled inrush	Wiring protection
2 TVS	Bidirectional, must not conduct at 60 V even at -40 °C	ISO 7637-2 fast pulses; bidirectional so a reversed battery does not turn it on
3 Front end	LM74800-Q1 + two 100 V N-MOSFETs, back to back	Reverse battery down to -65 V, output cut-off just above 60 V, inrush slew control (about 3 A expected without it)
4 Filter	Pi filter (C-L-C), optional CMC, bulk and hold-up capacitors	Conducted emissions (CISPR 25) and short supply dropouts
5 Main buck	LMR38020SDDAR, 4.2 to 80 V, 2 A (LCSC C3192337)	5 V rail for the CAN transceivers and the 3.3 V stage
6 3.3 V buck	From 5 V, about 1 A	MCU, PHY, displays, RFID, GNSS, flash
7 Modem buck	3.8 V, at least 3 A, own input from the protected rail	EC200A VBAT; transmit bursts stay off the MCU rail

A TVS alone is not enough: ISO 16750-2 asks for 10 load-dump pulses one minute apart and a TVS degrades with each one, so the front end disconnects instead of absorbing. The window between the 60 V target and the 80 V rating of the buck is too small for a TVS clamp, which is why the LM74800-Q1 cut-off sits in front of the buck.

Cranking: the 24 V supply range of ISO 16750-2 code E is 10 to 32 V, and momentary drops to about 9 V are a typical test. The buck regulates down to 4.2 V input at almost 100 % duty, so 5 V holds until the input falls to about 5.5 to 6 V. No boost stage is needed on a 24 V truck. Hold-up for short dropouts follows $C = 2Pt / (V1^2 - V2^2)$: 12 W for 10 ms from 24 V to 8 V needs about 470 µF.

Estimated load, to be measured on the prototype:

Rail	Load	Estimate	Basis
3.3 V	STM32H753 at 480 MHz	about 0.3 A	Absolute maximum is 620 mA on all VDD pins; refine with the CubeMX power calculator

Rail	Load	Estimate	Basis
3.3 V	PHY, ST7796 with backlight, MFRC522 with field on	about 0.1 A each	Confirm in each data sheet
3.3 V	GNSS with active antenna, OLED, touch, flash, codec, LEDs	about 0.2 A	Estimate
3.3 V	Total	about 0.8 A; budget 1 A	-
5 V	Two CAN transceivers, dominant state	about 0.15 A	Estimate
Own feed	ESP32-S3 10-inch module	0.8 to 1 A at 5 V (4 to 5 W)	Not measured; similar Waveshare panels draw about 4 W
3.8 V	Quectel EC200A	Idle about 20 mA; supply rated 2.8 A or more	Quectel reference design

Total: about 10 W of load, about 12 W from the vehicle with losses; about 0.5 A at 24 V and 1.3 A at 9 V during cranking, plus modem bursts.

I/O protection

Every line that leaves the board is protected at its connector, sized for a 24 V vehicle.

Interface	Protection	Note
CAN x2	TCAN1044AV-Q1 (±58 V bus fault) + ESD2CAN24-Q1 at the connector + split termination (2 x 60 Ω with a center capacitor)	Survives a short to the battery and the suppressed load dump; TCAN1042H variants reach ±70 V
100BASE-T1	CMC + ESD + DC-block capacitors of the 100 V class	Layout per TI SNLA434
SPI to ESP32-S3, 30 to 40 cm	1 to 4 MHz, low GPIO slew rate, 22 to 47 Ω series resistor at each driver, shielded cable with ground between signals, ESD array at both ends, CRC in each frame	Only data crosses the cable; the ESP32-S3 renders its own screen
I2C3, 1-Wire, RFID and LCD cables	Low-capacitance ESD array at each connector, series resistors	-
Vehicle inputs (ignition, wake)	Divider + RC + zener or TVS to ground	Never a diode clamp to 3.3 V, which back-feeds the rail
Local buttons	Series resistor + RC debounce + ESD array	-
Analog inputs	Series resistor + capacitor (anti-alias) + low-leakage clamp; divider for 24 V sensors	-
SPI chip selects	10 k pull-up to 3.3 V on RFID_CS (PA0), LCD_CS (PA4) and SPI4_CS (PE0)	GPIOs leave reset in analog mode and float; the pull-ups keep every slave deselected while PB3 and PA15 still sit in their debug state during boot

Layout rule: the surge must meet the protection first. The trace runs connector, TVS pad, series resistor, then the IC, with no stub and no via before the TVS; the TVS ground drops straight into the plane with several vias next to the connector ground pin. An 8 kV ESD strike peaks near 30 A in about 1 ns, so 10 nH of trace and via between connector and TVS add 300 V on top of the clamp voltage.

Standards: ISO 16750-2 (supply), ISO 7637-2 and -3 (transients), ISO 10605 (ESD), CISPR 25 (emissions), ISO 11452 (immunity); parts AEC-Q100, Q101 and Q200 where available.

Connectors and harnesses

Five connectors group the cables by destination: two go to the vehicle harness, two to the panel, and one keeps the free pins for expansion. For the prototype every family is cheap and needs no crimping: solder cups, a ready-made patch cord or IDC ribbon pressed in a vise. Pin numbers are a proposal; the layout may reorder them as long as pairs stay together and each fast signal keeps a ground next to it.

Ref	Function	Family and parts	Ways	Plate
X1	Power + vehicle I/O	DB15 female right angle on the board: XKB X0315FT17B2H (LCSC C18723073); cable side DB15 male with solder cups (code to confirm)	15	Vehicle
X2	CAN x2 + T1	RJ45 8P8C right angle, shielded, without magnetics: kinghelm KH-RJ45-56-8P8C-D (C2828082) or Ckmtw R-RJ45R08P-8000 (C386760); confirm "no magnetics" in the data sheet; ready-made Cat5e patch cord	8	Vehicle
X3	ESP32-S3 link	IDC box header 2x8, 2.54 mm, right angle; IDC socket and 16-way ribbon (codes to confirm)	16	Panel
X4	HMI panel	IDC box header 2x20, 2.54 mm, right angle: TXGA FBH25402-D40B1001K6K (C2834113); IDC socket and 40-way ribbon	40	Panel
X5	Expansion	2x10 box header, 2.54 mm, vertical	20	Inside
J_DBG	Debug + console	STDC14 2x7 1.27 mm, Samtec FTSH-107-01-L-DV-K	14	Panel
J_RF1-3	LTE main, LTE diversity, GNSS	u.FL on the board, pigtail to an SMA bulkhead jack	2 each	Vehicle

Every connector carries at least two ground pins; all except X2 also carry a supply pin (an 8-way RJ45 has no room left after three pairs and ground). IDC contacts carry about 1 A, so the 5 V lines of X3 and X4 use two contacts each. Mixing up X3 and X4 is impossible: they have different sizes.

X1 - Power + vehicle I/O (DB15)

Pin	Signal	MCU / net	Note
1	24V_IN (KL30)	-	Fuse, TVS, LM74800-Q1
2	24V_IN (KL30)	-	Second contact shares the current
3	GND (KL31)	-	To the ground plane at the connector
4	GND (KL31)	-	-
5	KL15_IN	PF13	ADC2_INP2; divider, RC, TVS; polled
6	AGND	-	Return for the analog sensors
7	AIN1	PC0	ADC123_INP10; series R, C, clamp
8	AIN2	PF3	ADC3_INP5
9	AIN3	PF4	ADC3_INP9
10	AIO4	PA5	ADC12_INP19 or DAC1_OUT2
11	5V_SENS	-	Current-limited 5 V for sensors and 1-Wire
12	1W_DATA	PD1	UART4_TX half-duplex; 4.7 k pull-up, ESD
13	GND	-	-

Pin	Signal	MCU / net	Note
14	spare	-	-
15	spare	-	-

X2 - Vehicle networks (RJ45, T568B pairs)

Pin	Signal	MCU / net	Note
1	CAN1_H	-	T568B pair 2 (pins 1-2); TCAN1044AV-Q1 #1, STB PD4
2	CAN1_L	-	-
3	CAN2_H	-	T568B pair 3 (pins 3-6); TCAN1044AV-Q1 #2, STB PD6
4	T1_P	-	T568B pair 1 (pins 4-5); DP83TC812 TRD_P via CMC
5	T1_N	-	-
6	CAN2_L	-	-
7	GND	-	T568B pair 4 (pins 7-8)
8	GND	-	-

X3 - ESP32-S3 link (IDC 2x8)

Pin	Signal	MCU / net	Note
1	5V_DISP	-	Own 5 V buck, about 1 A
2	5V_DISP	-	Two IDC contacts for the current
3	GND	-	-
4	SCK	PE12	SPI4; 22-47 ohm series at the STM32
5	GND	-	-
6	MOSI	PE14	Series resistor at the STM32
7	GND	-	-
8	MISO	PE13	Series resistor at the ESP32
9	GND	-	-
10	CS	PE0	Software chip select
11	IRQ	PD14	Data ready from the ESP32 (EXTI14)
12	EN	PD13	ESP32 reset/enable
13	GND	-	-
14	GND	-	-
15	spare	-	-
16	spare	-	-

X4 - HMI panel (IDC 2x20)

Pin	Signal	MCU / net	Note
1	5V	-	Panel supply

Pin	Signal	MCU / net	Note
2	5V	-	-
3	3V3	-	Button common (inputs use pull-down)
4	GND	-	-
5	LCD_SCK	PD3	SPI2
6	GND	-	Next to SCK
7	LCD_MOSI	PC3	SPI2
8	LCD_MISO	PC2	SPI2
9	LCD_CS	PA4	-
10	LCD_DC	PA3	-
11	LCD_RST	PD7	-
12	LCD_BL	PE15	On/off only
13	GND	-	-
14	I2C2_SDA	PF0	Touch + OLED
15	I2C2_SCL	PF1	-
16	TOUCH_INT	PG8	FT6336
17	TOUCH_RST	PG5	-
18	GND	-	-
19	RFID_SCK	PB3	SPI1
20	GND	-	Next to SCK
21	RFID_MOSI	PB5	SPI1
22	RFID_MISO	PB4	SPI1
23	RFID_CS	PA0	-
24	RFID_RST	PG0	-
25	RFID_IRQ	PG1	Polled
26	BTN1	PB1	-
27	BTN2	PD0	-
28	BTN3	PG4	-
29	BTN4	PD5	-
30	BTN5	PE6	-
31	BTN6	PE3	-
32	BTN7	PD12	-
33	BTN8	PF9	-
34	BTN9	PD15	-

Pin	Signal	MCU / net	Note
35	BTN10	PE11	-
36	I2C3_SCL	PA8	Panel expansion
37	I2C3_SDA	PC9	-
38	GND	-	-
39	SPK+	-	From the codec / amplifier
40	SPK-	-	-

X5 - Expansion (2x10 box header)

Pin	Signal	MCU / net	Note
1	3V3	-	-
2	5V	-	-
3	GND	-	-
4	GPIO	PA6	Polled GPIO, ADC or PWM
5	GPIO	PB10	Polled GPIO, ADC or PWM
6	GPIO	PB11	Polled GPIO, ADC or PWM
7	GPIO	PB15	Polled GPIO, ADC or PWM
8	GPIO	PC6	Polled GPIO, ADC or PWM
9	GPIO	PC8	Polled GPIO, ADC or PWM
10	GPIO	PD2	Polled GPIO, ADC or PWM
11	GPIO	PD11	Polled GPIO, ADC or PWM
12	GPIO	PF5	Polled GPIO, ADC or PWM
13	GPIO	PF8	Polled GPIO, ADC or PWM
14	GPIO	PF14	Polled GPIO, ADC or PWM
15	GPIO	PF15	Polled GPIO, ADC or PWM
16	GPIO	PG6	Polled GPIO, ADC or PWM
17	GPIO	PG12	Polled GPIO, ADC or PWM
18	GND	-	-
19	GND	-	-
20	GND	-	-

J_DBG - STDC14 debug and console (pin numbers per ST UM2448):

Signal	MCU / net	Note
SWDIO	PA13	Debug and flash
SWCLK	PA14	-
NRST	NRST	Pin 25

Signal	MCU / net	Note
VCP_TX / VCP_RX	PD8 / PD9	USART3 console; TX/RX named from the probe side
VDD sense	3V3	Reference for the probe
GND	net	Several pins
SWO, JTAG	not wired	PB3, PA15, PB4 are in use

J_RF - antennas: RF1 LTE main, RF2 LTE diversity (optional), RF3 GNSS with DC bias for an active antenna; u.FL on the board, pigtail to an SMA bulkhead on the vehicle end plate.

Harnesses:

Harness	Cable	Notes
X1	DB15 male with solder cups; 20 AWG (about 0.5 mm²) on 24 V and ground	Two contacts each for 24 V and ground; hood with strain relief
X2	Ready-made Cat5e patch cord, cut at the vehicle end	T1 on pair 4-5, CAN1 on 1-2, CAN2 on 3-6, ground on 7-8; label "not PC Ethernet"
X3	16-way ribbon, 30 to 40 cm	Ground between fast signals by pin order; SPI at 1 to 4 MHz
X4	40-way ribbon, about 36 cm	Ground next to each SCK; expect a lower frame rate on the ST7796
X5	Internal only	Polled GPIO, ADC or PWM

Pin map

49 pins are used by the current firmware, 45 are planned, and 6 Nucleo-only pins are avoided (pin map rev27); every pin and AF resolves to a macro in the port pinmap.h and matches CubeMX.

In use today:

Function	Pins	Peripheral
Ethernet RMII	PA1, PA2, PA7, PB13, PC1, PC4, PC5, PG11, PG13	ETH, AF11
ST7796 display	PC2, PC3, PD3; PA3 DC, PA4 CS, PD7 RESET, PE15 LED	SPI2 AF5 + GPIO
FT6336 touch	PF0, PF1; PG5 RST, PG8 INT	I2C2 AF4 + GPIO
SSD1306 OLED	PF0, PF1	I2C2 AF4
MFRC522 RFID	PB3, PB4, PB5; PA0 CS	SPI1 AF5 + GPIO
FDCAN1	PB8 RX, PB9 TX	AF9
Console	PD8 TX, PD9 RX	USART3 AF7
LEDs	PB0, PE1, PB14	GPIO
Buttons	PC13 (USER), PB1, PD0, PG4, PD5, PE6, PE3, PD12, PF9, PD15, PE11	GPIO + EXTI
SWD	PA13, PA14	Debug
HSE	PH0, PH1	Oscillator
DEV_GPIO	PE5 (INT1); PE2, PE4 defined only	GPIO; all three avoided

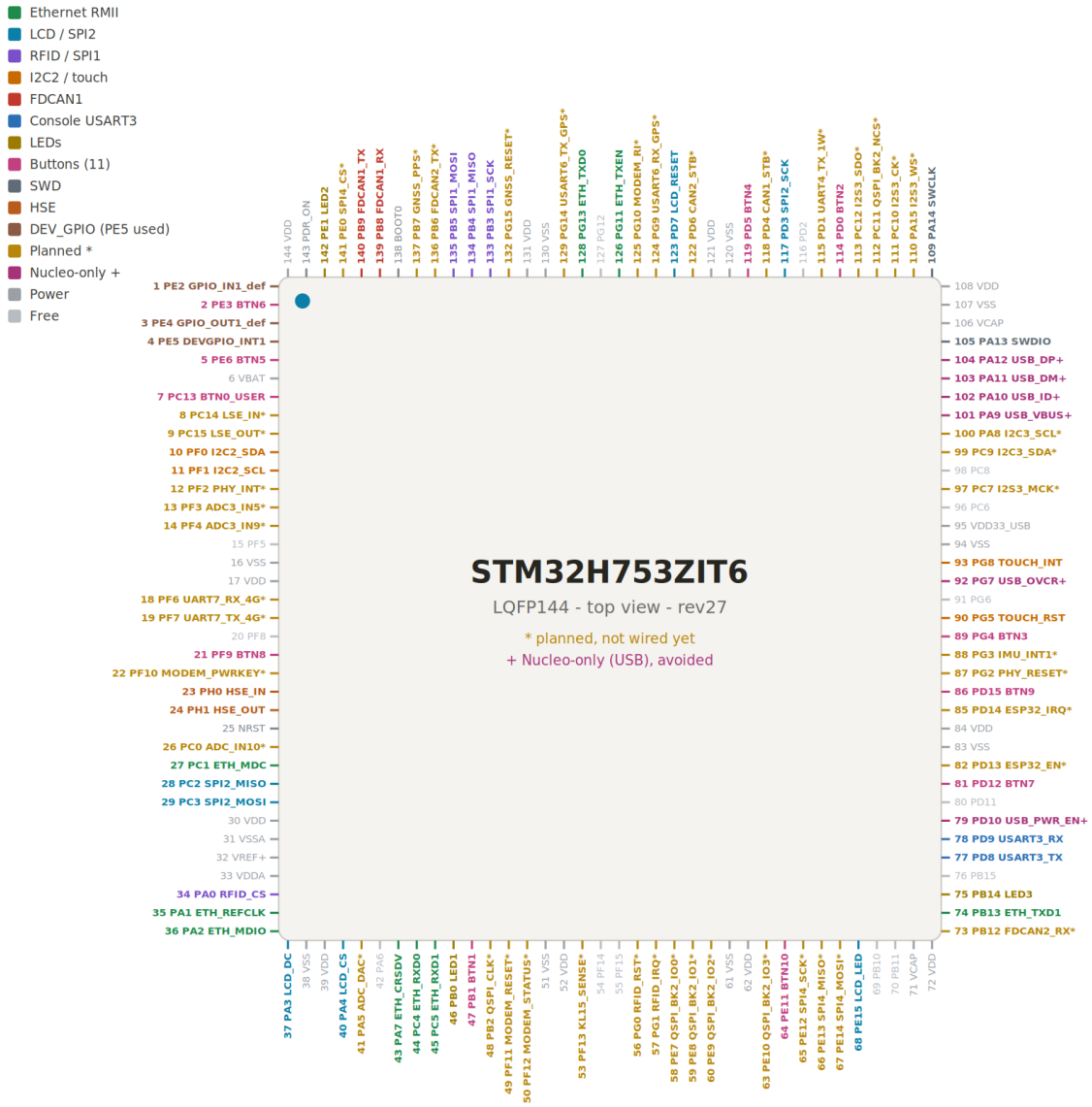
Planned in rev27:

Function	Pins	Peripheral
Audio	PC10 CK, PA15 WS, PC12 SDO, PC7 MCK	I2S3 AF6
ESP32-S3 link	PE12 SCK, PE13 MISO, PE14 MOSI; PE0 CS	SPI4 AF5 + GPIO
QSPI flash	PB2 CLK, PC11 NCS, PE7 to PE10 IO0 to IO3	QUADSPI bank 2, AF9/AF10
4G modem	PF7 TX, PF6 RX	UART7 AF7
GNSS	PG14 TX, PG9 RX	USART6 AF7
1-Wire	PD1	UART4 AF8, half-duplex
FDCAN2	PB6 TX, PB12 RX	AF9
Expansion I2C	PA8 SCL, PC9 SDA	I2C3 AF4, shared with the IMU
IMU	PG3 INT1	GPIO, polled
Analog	PC0, PF3, PF4; PA5	ADC; ADC or DAC1_OUT2
PHY control	PG2 RESET, PF2 INT/wake	GPIO; INT on EXTI2
ESP32-S3 control	PD14 IRQ, PD13 EN	GPIO; IRQ on EXTI14
Modem control	PF10 PWRKEY, PF11 RESET, PF12 STATUS, PG10 RI	GPIO; RI on EXTI10
GNSS control	PG15 RESET, PB7 PPS	GPIO; PPS on EXTI7
CAN standby	PD4 (CAN1), PD6 (CAN2)	GPIO

Function	Pins	Peripheral
LSE	PC14, PC15	OSC32
Ignition input	PF13 KL15	ADC2_INP2, polled
RFID control	PG0 RST, PG1 IRQ	GPIO; IRQ polled
Expansion (X5)	PA6, PB10, PB11, PB15, PC6, PC8, PD2, PD11, PF5, PF8, PF14, PF15, PG6, PG12	Free GPIOs, no EXTI line
Avoided	PA9 to PA12, PD10, PG7	Nucleo USB (UM2407 Table 12)

LQFP144 pinout

The package view shows each pin with its function; amber pins are planned and not wired yet.



stm32h753zi_lqfp144_pinout_v8 · top view

Pin map review and release gate

An independent cross-check of every peripheral against the ST data sheet DS12117 (Rev. 10) for the LQFP144 package found all pin and alternate-function assignments correct; four points remain before the layout is released.

Peripheral	Pins	Result
Ethernet RMII	PA1, PA2, PA7, PB13, PC1, PC4, PC5, PG11, PG13	OK
FDCAN1 / FDCAN2	PB8, PB9 / PB6, PB12	OK
SPI1 (RFID) / SPI2 (ST7796) / SPI4 (ESP32-S3)	PB3, PB4, PB5 / PD3, PC2, PC3 / PE12, PE13, PE14	OK
QSPI bank 2	PB2, PC11, PE7 to PE10	OK
UART7 / USART6 / UART4	PF6, PF7 / PG9, PG14 / PD1	OK
I2C2 / I2C3	PF0, PF1 / PA8, PC9	OK
I2S3	PC10, PA15, PC12, PC7	OK
ADC / DAC	PC0, PF3, PF4, PA5, PF13	OK

Point	Impact	Resolution	Status
EXTI5 shared by BTN4 (PD5) and GPIO_INT1 (PE5)	Only one port can drive an EXTI line; arming GPIO_INT1 silently kills BTN4, and the button driver only checks buttons against each other	BTN4 keeps its interrupt. On the custom board GPIO_INT1 is removed (or CONFIG_DEV_GPIO disabled); PE5 stays unused	Decided; board.h change pending
EXTI3 shared by BTN6 (PE3) and IMU INT1 (PG3)	Same EXTI rule	IMU INT1 is polled (status register or FIFO watermark); BTN6 keeps its interrupt	Decided
Debug pins PA15, PB3, PB4 reused by I2S3 and SPI1	JTAG and the SWO trace cannot be used once the firmware reconfigures these pins	SWD only (PA13, PA14) on the STDC14; select SWD and turn SWO off in the debugger; console on USART3 (VCP) instead of SWO; 10 k pull-ups on the SPI chip selects for the boot window	Decided
NuttX port: SPI3 labels	Kconfig "pin set 1/2" labels are swapped versus board.h	Fix the Kconfig help text; keep SPI3 disabled here (the block is I2S3)	Port fix pending
NuttX port: SPI5 pin set 1	MOSI resolves to PF11, which is MODEM_RESET in this design	Keep SPI5 disabled; fix or remove the pin set in board.h and Kconfig	Port fix pending
NuttX port: OTGFS_PWRON	Defined on PG6 instead of PD10; PG6 is an expansion pin on X5 and would be driven if USB host is enabled	Move to PD10 (MB1364, SB77); USB stays out of rev A	Port fix pending

Release gate: the layout starts after the three port fixes and the GPIO_INT1 removal are in board.h and Kconfig. This review covers the logical pin map only; the schematic (ERC, the TI DP83TC812 review checklist, protection values) and the routing (DRC, impedance, return paths) are reviewed separately.

Power pins

All supply pins run from one 3.3 V rail; the core voltage comes from the internal LDO through the two VCAP pins.

Rail	Pins	Notes
VDD	17, 30, 39, 52, 62, 72, 84, 108, 121, 131, 144	3.3 V; decoupling per AN4938
VSS	16, 38, 51, 61, 83, 94, 107, 120, 130	Ground plane
VDDA / VSSA	33 / 31	Analog supply, filtered from 3.3 V
VREF+	32	ADC and DAC reference
VBAT	6	Backup domain; tie to VDD if no battery
VCAP	71, 106	Internal LDO output; capacitor values per DS12117
VDD33_USB	95	USB unused; connect per AN4938
NRST, BOOT0, PDR_ON	25, 138, 143	Reset, boot select, power-down reset enable

Reference documents: AN4938 (hardware development), DS12117 (data sheet) and RM0433 (reference manual).

Firmware architecture

One NuttX board configuration exposes every resource as a standard device, so the application never touches registers.

Resource	NuttX side	Notes
Console	USART3, NSH	Reached through the STDC14 VCP of the external ST-LINK-V3
Ethernet	eth0, stm32h7 MAC driver	DP83TC812 init over MDIO to write, following TI SNLA404
CAN	FDCAN1, FDCAN2	Character driver or SocketCAN
Displays	ST7796 framebuffer + LVGL; SSD1306 /dev/lcd0	FT6336 as touch input device
RFID	MFRC522 driver on SPI1	-
Buttons	/dev/buttons	11 inputs
Flash	w25qxxxjv MTD + LittleFS	CONFIG_STM32_QSPI_MODE_BANK2
4G and GNSS	UART7, USART6 serial ports; control GPIOs	PPP or AT client; NMEA parser
1-Wire	UART4 lower-half	Driver must be ported to the H7 USART
Audio	I2S3 + codec driver	Codec not chosen
Analog	ADC and DAC drivers	-
IMU	LSM6DSOX on I2C3, uORB	Port from the upstream lsm6dso32 uORB driver: full-scale table and self-test presets differ
ESP32-S3 link	SPI4 master + IRQ on PD14	vrm_net (or a SPI derivative) with CRC and retransmit
OTA	Image staged in QSPI flash	Bootloader not chosen (MCUboot is an option)

Resource	NuttX side	Notes
Network layer	vrm_net over UDP	Custom data-centric protocol

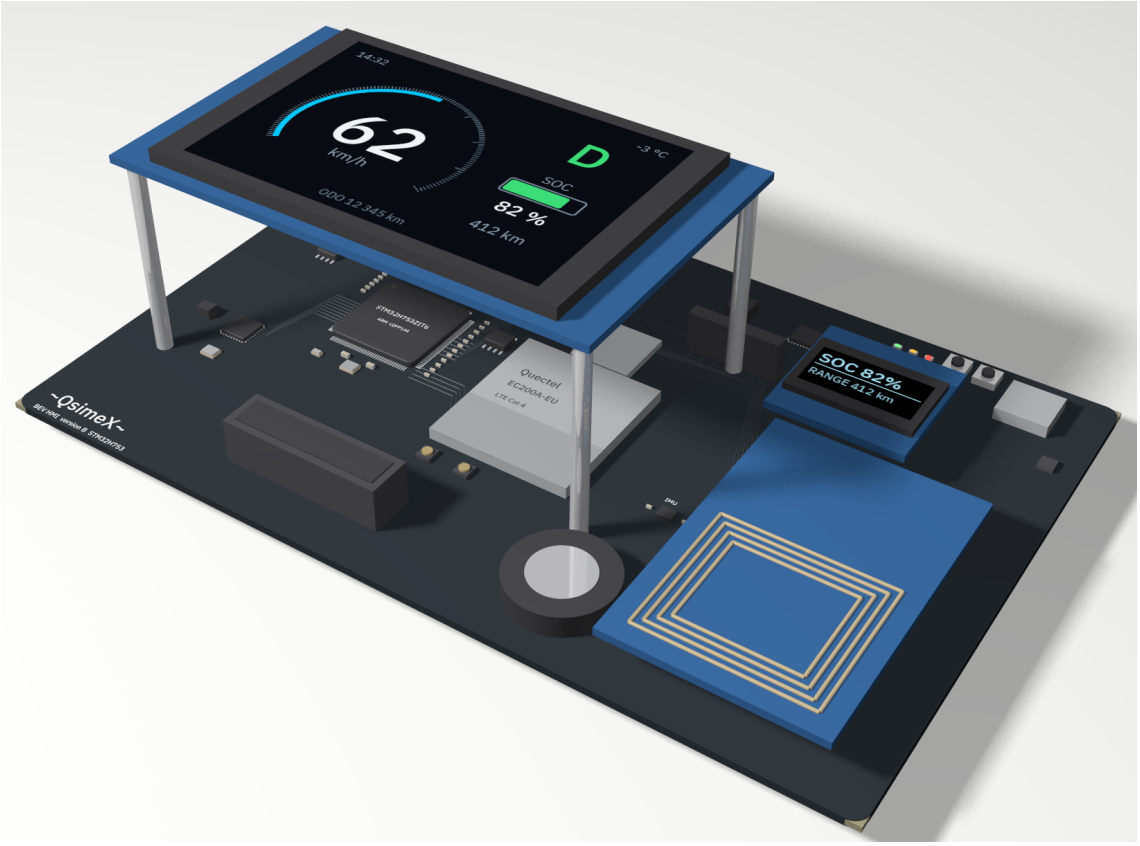
Version B: integrated HMI

Version B puts the ST7796 display with its touch panel, the OLED, the RFID reader and the speaker on the board itself, so the ECU becomes the HMI unit. The pin map does not change; only the content of X4 and the mechanics change.

Item	Version A	Version B
ST7796 + FT6336 touch	External, through the X4 ribbon	On the board: module on 11 mm standoffs above the electronics, SPI2 on short traces
SSD1306 OLED	External, on X4	On the board, on a 4-way header
RFID MFRC522	External, on X4	On the board (RC522-class module) behind a tap zone on the front panel
Speaker	In the panel case, on X4	On the board, behind a grille
IMU LSM6DSOX	On the board	On the board
Buttons	External, on X4	External, on X4 reduced to 14 ways
X4	IDC 2x20, 40 ways	IDC 2x7, 14 ways: 10 buttons, 3V3, 3 GND
X1, X2, X3, X5, debug, RF	As described above	Same pinout; vertical parts on the bottom side, exiting through the rear cover
Board	110 x 130 mm	About 170 x 110 mm
Enclosure	Extruded aluminium with two end plates	ABS pod, about 190 x 130 x 48 mm; a metal front would block the 13.56 MHz RFID field

Design notes for version B:

- RFID antenna: no copper pour on any layer under or around the antenna loop, no metal in front of it, and the reader at least a few centimetres away from the speaker magnet.
- Display SPI: with SPI2 on short traces the ST7796 can run at a higher clock than through the ribbon of version A.
- Heat: ABS does not spread heat like the aluminium profile. The modem, the bucks and the MCU need copper pours and vent slots in the pod.
- IMU: rigidly mounted near the board centre and near a fixing point, away from the bucks and the speaker, so vibration and heat do not bias it.
- Ground: without a metal enclosure there is no chassis shield; the ESD and filter parts at the rear connectors carry all the protection.



version B board \u00b7 display lifted 30 mm (exploded view) to show the electronics below \u00b7 POV-Ray



version B pod \u00b7 front with display, OLED, RFID tap zone and speaker \u00b7 rear with X1, X2, audio, X3, X4, debug and three SMA \u00b7 same scale

X4 in version B - Buttons (IDC 2x7)

Pin	Signal	MCU / net	Note
1	3V3	-	Button common (inputs use pull-down)
2	BTN1	PB1	-
3	BTN2	PD0	-
4	BTN3	PG4	-
5	BTN4	PD5	-
6	BTN5	PE6	-
7	GND	-	-

Pin	Signal	MCU / net	Note
8	BTN6	PE3	-
9	BTN7	PD12	-
10	BTN8	PF9	-
11	BTN9	PD15	-
12	BTN10	PE11	-
13	GND	-	-
14	GND	-	-

Risks and open decisions

The largest open points are the T1 bring-up on NuttX, four parts that are still unselected, and a few pin-set fixes in the board port.

Item	Why it matters	Next step
T1 link role (master or slave)	Sets a PHY strap	Decide per topology; bench default is board = slave
DP83TC812 driver	No NuttX support today	Port the TI init sequence; test with a T1 media converter
Bench access to a PC	T1 cannot plug into a PC	Buy a 100BASE-T1 to 100BASE-TX converter
QSPI bank 2	First use of this mode on the board	Bring up w25qxxxjv early on a Nucleo with flying wires
1-Wire on H7	Upstream lower-half targets F1/F4	Port or drop before layout freeze
Unchosen parts	GNSS module, codec, TVS and MOSFETs of the front end, modem buck (3 A or more)	Pick parts with LCSC stock and NuttX drivers
Load dump class	Front end sized for the suppressed 58 V pulse	If the truck has no central suppression (151 to 202 V), raise TVS and MOSFET ratings; the LM74800-Q1 topology stays
10-inch display power	Not measured; estimated 4 to 5 W	Measure; feed it from its own buck, not from the 2 A 5 V rail
EXTI budget	All 16 EXTI lines used; EXTI5 (BTN4 / GPIO_INT1) and EXTI3 (BTN6 / IMU INT1) are shared	GPIO_INT1 removed, IMU INT1 polled (see the release gate); new interrupt inputs must be polled
RF coexistence	LTE transmit near GNSS	Separate edges, filtered supplies; measure on prototype
Port board.h pin sets	SPI3 Kconfig labels swapped vs board.h; SPI5 set 1 MOSI is PF11; port H I2C sets not in LQFP144; OTGFS_PWRON on PG6 instead of PD10	Release gate: fix board.h and Kconfig before the layout; wire by board.h until then
Pin budget	14 GPIOs left, all on X5, none with a free EXTI line	Next interrupt input needs polling, a freed line or LQFP176
Connector codes	DB15 male solder plug, IDC sockets and 2x8 header not confirmed on LCSC; RJ45 must be without magnetics	Check the codes and the RJ45 data sheet before ordering
Display SPI on X4	SPI2 to the ST7796 runs single-ended through the panel cable	Expect a lower frame rate; series resistors, ribbon with GND next to SCK
USB service port	No USB in rev A; field flashing and console need the probe	Decide whether to add native USB-C (DFU + CDC-ACM) on PA11/PA12 before the board outline

Backlog

Kept out of rev A on purpose.

- microSD on SDMMC1: blocked by I2S3 (PC10/PC12) and I2C3 (PC9) in LQFP144.

- Native 7-inch RGB display with LTDC and 16-bit SDRAM: needs LQFP176/208.
- Gigabit 1000BASE-T1: needs an MCU with RGMII; the PHY footprint already fits DP83TG720.
- USB device or host on PA9 to PA12.
- Rotary encoder on a timer: needs a free CH1 + CH2 pair.